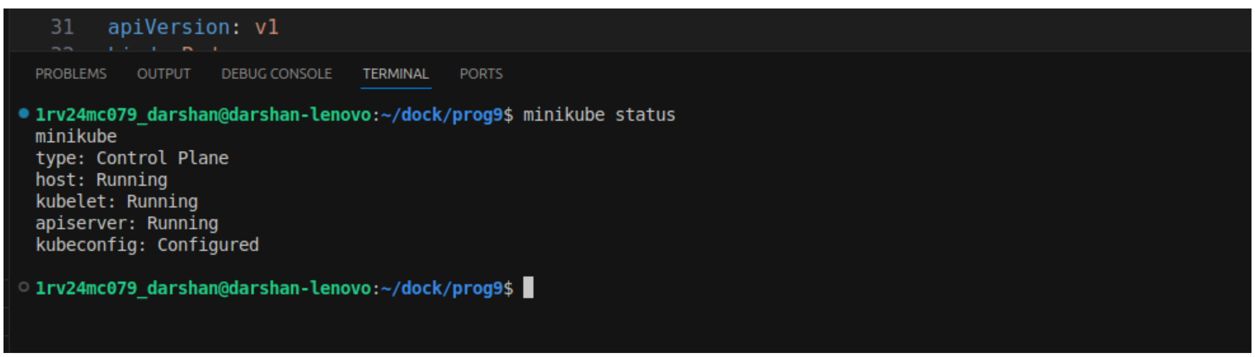
Program 9:
Organize Kubernetes Pods using Labels and Namespace

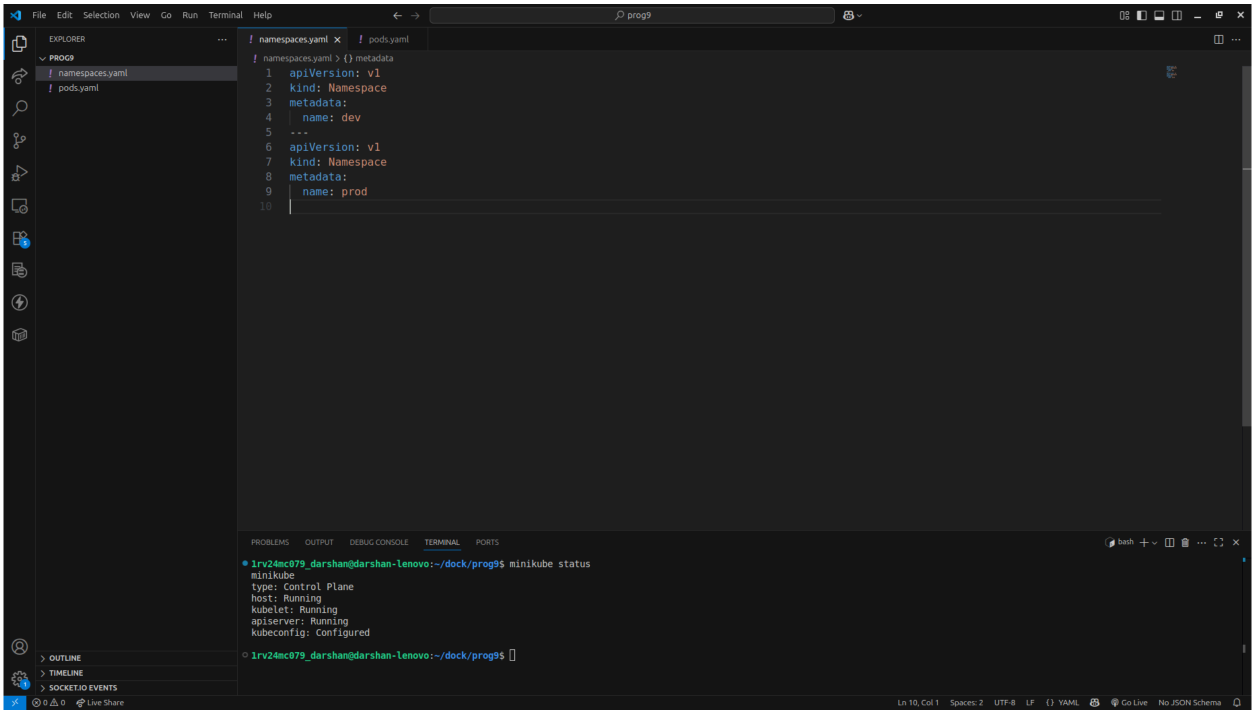
Project Structure:
program_9/
—>namespaces.yaml
—->pods.yaml

Procedure:

Create a folder (ex: program_9)
Step 1: Start Minikube
minikube start
Check the status of Minikube
minikube status

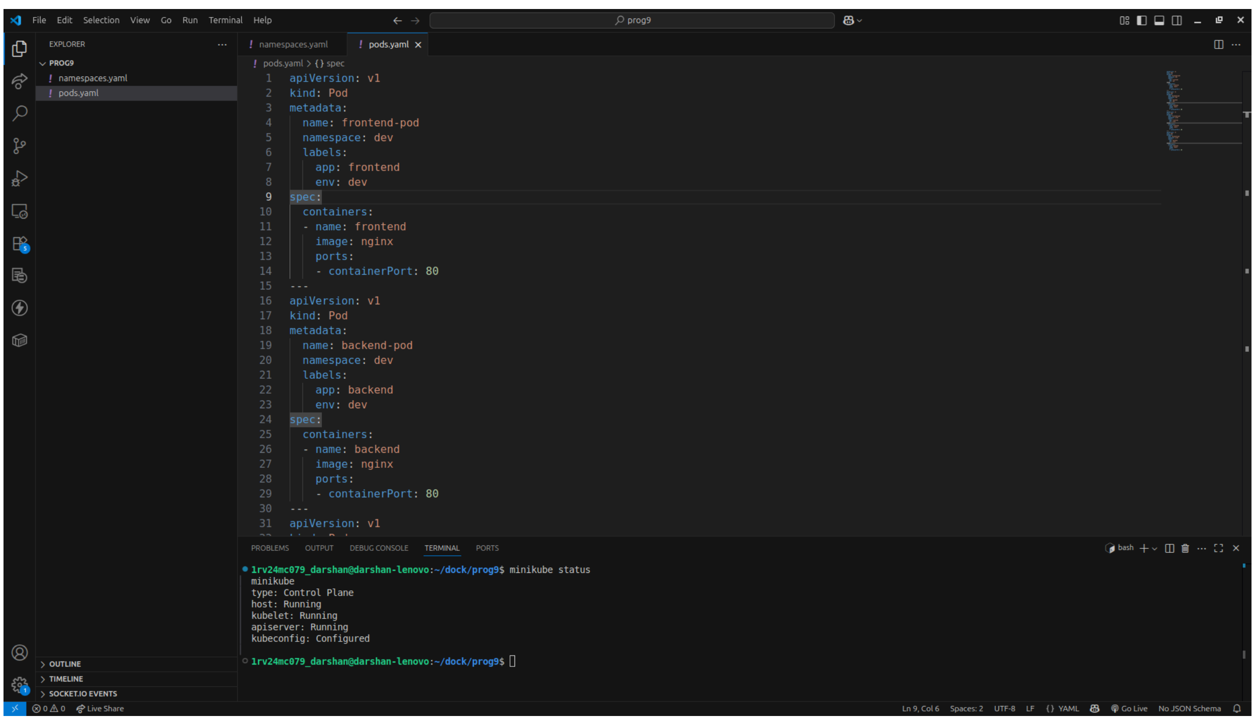


Step 2: Create namespaces.yaml file with the following code
nano namespaces.yaml



Step 3: Create pods.yaml file with the following code

nano pods.yaml



Step 4: Apply the files

kubectl apply -f namespaces.yaml

kubectl apply -f pods.yaml

```
31   apiVersion: v1

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$
```

Step 5: Verify the pods with namespaces

kubectl get pods -n dev

kubectl get pods -n prod

```
31   apiVersion: v1

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-65d889d5b8-2p7jg   1/1     Running   0          68s
nginx-deployment-65d889d5b8-mnhp9   1/1     Running   0          68s
1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   2/2     2            2           6m15s
1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl get deployments.apps
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   2/2     2            2           6m17s
1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$
```

Step 6: Get all of the pods

docker get pods -A

Step 7: Verify pods with labels

kubectl get pods -n dev -l app=frontend

kubectl get pods -n dev -l app=backend

kubectl get pods -n prod -l app=frontend

kubectl get pods -n prod -l app=backend

```
31   apiVersion: v1

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl get pods -n dev -l app=frontend
NAME           READY   STATUS              RESTARTS   AGE
frontend-pod   0/1     ContainerCreating   0          51s
1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl get pods -n dev -l app=backend
NAME          READY   STATUS              RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          57s
1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$
```